

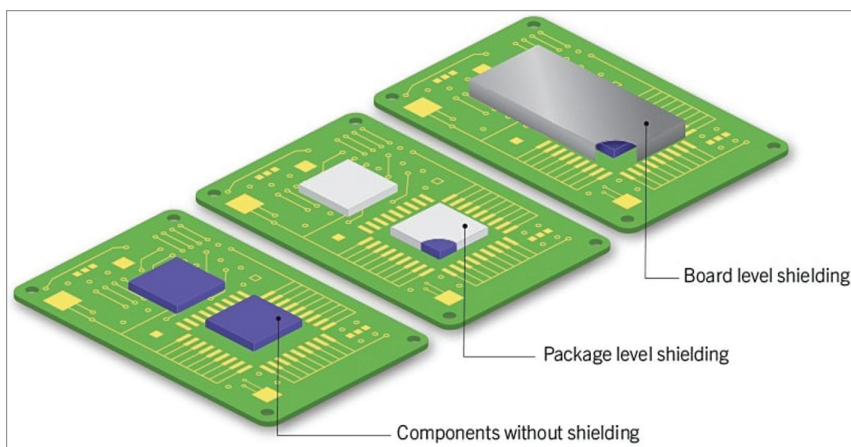


Fig. 4: Depiction of package-level and board-level EMI shielding



Fig. 5: Depiction of common automotive potting uses, including encapsulating stators (left) and as a thermal interface material for battery packs (right)

coating fillers.

Unlike conformal coatings, which are applied directly to the circuitry, conductive coatings are typically used on the housing units along the outer perimeter; however, there are instances where specific components require shielding (known as package-level shielding). Fig. 4 illustrates the different shielding types needed to achieve electromagnetic compliance.

Enhancing protection using potting resins

Conformal coatings provide environmental protection, while potting and encapsulating resins extend this protection to physical impacts. Materials such as epoxy, polyurethane, and silicone resins are used to encase circuit boards and sensors within housing units. Epoxy resins cure to a stiff, rigid finish and provide the most protection from physical impacts and exposure to harsh chemicals. Furthermore, their excellent dielectric properties make them an ideal resin choice for

encapsulating electric motor stators, grommets, and relays, helping prevent electrical discharge and shorts. Low-permittivity resins are suitable for encapsulating receivers for chargers, controllers, and cable harnesses, thereby ensuring high signal fidelity.

Thermally conductive potting compounds are flowable materials loaded with ceramic fillers that function like a thermally conductive adhesive. The difference between the two product types is that adhesives are applied locally, whereas a potting compound is intended to fill volume. Some are designed to help dissipate heat around battery packs and modules, while others are flame-retardant compounds that can help delay the spread of fire in battery systems.

Don't forget the grease!

Grease is a semi-solid material primarily used to reduce friction and wear between moving parts. Beyond lubrication, speciality greases play an important role in thermal



Fig. 6: Illustration of proper use of dielectric grease (on top of the terminal) compared to an electrically conductive grease, which is being used as a gap filler to improve current flow from the terminal to the clamp

regulation, electrical grounding, and corrosion prevention. Based on several case studies, our electrically conductive greases are commonly used as makeshift electrical gap fillers and as dual-purpose lubricants for electrical contacts. Thicker, electrically conductive pastes are ideal for static applications that do not involve moving parts, such as helping bridge the power inverter to the electric motor. Other greases combine electrical conductivity with lubrication and can be used to ground stray currents from inverters, preventing corrosion of moving parts like belts or bearings.

Like our thermal adhesives, our thermal grease helps prevent components in various systems from overheating. The advantage of thermal paste is its ease of rework, as worn processors can be quickly accessed and replaced if necessary.

Dielectric grease is an electrically insulating compound, yet it is misunderstood in terms of its use. Its intended use is as a hydrophobic barrier on electrical connectors to prevent corrosion; circuit housing units, such as those for motors on e-bikes, use dielectric grease to seal around the seams, preventing exposure to moisture and corrosive elements. Dielectric grease is also suitable for use on car battery terminals; however, users must take care to ensure it's not misused. Fig. 6 illustrates the correct use of dielectric grease compared to electrically conductive grease. **EFY**